

```

1 // Partie commune
2
3 public class Record {
4     String INE;    String mathematics;    String middleware;    String networks;
5     String systems;    String architecture;    String programming;
6
7     public Record() {}
8
9     public Record(String INE, String mathematics, String middleware, String networks,
10        String systems, String architecture, String programming) {
11         super();
12         this.INE = INE;
13         this.mathematics = mathematics;
14         this.middleware = middleware;
15         this.networks = networks;
16         this.systems = systems;
17         this.architecture = architecture;
18         this.programming = programming;
19     }
20
21     public String getINE() { return INE; }
22     public void setINE(String INE) { this.INE = INE; }
23     public String getMathematics() { return mathematics; }
24     public void setMathematics(String mathematics) { this.mathematics = mathematics; }
25     public String getMiddleware() { return middleware; }
26     public void setMiddleware(String middleware) { this.middleware = middleware; }
27     public String getNetworks() { return networks; }
28     public void setNetworks(String networks) { this.networks = networks; }
29     public String getSystems() { return systems; }
30     public void setSystems(String systems) { this.systems = systems; }
31     public String getArchitecture() { return architecture; }
32     public void setArchitecture(String architecture) { this.architecture = architecture; }
33     public String getProgramming() { return programming; }
34     public void setProgramming(String programming) { this.programming = programming; }
35 }
36
37 public class Student {
38     String INE;    String firstname;    String lastname;    String birthdate;
39     String sex;    String address;    String city;    String zip;
40     String country;    String phone;    String email;
41
42     public Student() {}
43
44     public Student(String INE, String firstname, String lastname,
45        String birthdate, String sex, String address, String city,
46        String zip, String country, String phone, String email) {
47         super();
48         this.INE = INE;
49         this.firstname = firstname;
50         this.lastname = lastname;
51         this.birthdate = birthdate;
52         this.sex = sex;
53         this.address = address;

```

```

54     this.city = city;
55     this.zip = zip;
56     this.country = country;
57     this.phone = phone;
58     this.email = email;
59 }
60
61 public String getINE() { return INE; }
62 public void setINE(String INE) { this.INE = INE; }
63 public String getFirstname() { return firstname; }
64 public void setFirstname(String firstname) { this.firstname = firstname; }
65 public String getLastname() { return lastname; }
66 public void setLastname(String lastname) { this.lastname = lastname; }
67 public String getBirthdate() { return birthdate; }
68 public void setBirthdate(String birthdate) { this.birthdate = birthdate; }
69 public String getSex() { return sex; }
70 public void setSex(String sex) { this.sex = sex; }
71 public String getAddress() { return address; }
72 public void setAddress(String address) { this.address = address; }
73 public String getCity() { return city; }
74 public void setCity(String city) { this.city = city; }
75 public String getZip() { return zip; }
76 public void setZip(String zip) { this.zip = zip; }
77 public String getCountry() { return country; }
78 public void setCountry(String country) { this.country = country; }
79 public String getPhone() { return phone; }
80 public void setPhone(String phone) { this.phone = phone; }
81 public String getEmail() { return email; }
82 public void setEmail(String email) { this.email = email; }
83 }
84
85 // Partie client
86
87 import jakarta.ws.rs.GET;
88 import jakarta.ws.rs.Produces;
89 import jakarta.ws.rs.QueryParam;
90 import jakarta.ws.rs.Path;
91
92 @Path("/")
93 public interface ServiceInterface {
94     @GET
95     @Path("/getstudent")
96     @Produces({"application/json"})
97     public Student getStudent(@QueryParam("firstname") String firstName,
98                               @QueryParam("lastname") String lastName);
99
100     @GET
101     @Path("/getrecord")
102     @Produces({"application/json"})
103     public Record getRecord(@QueryParam("ine") String ine);
104 }
105
106 import jakarta.ws.rs.client.ClientBuilder;
107 import jakarta.ws.rs.core.UriBuilder;

```

```

108 import org.jboss.resteasy.client.jaxrs.ResteasyClient;
109 import org.jboss.resteasy.client.jaxrs.ResteasyClientBuilder;
110 import org.jboss.resteasy.client.jaxrs.ResteasyWebTarget;
111 import org.jboss.resteasy.client.jaxrs.internal.ResteasyClientBuilderImpl;
112
113 public class TestClient {
114     public static final String url = "http://localhost:8080";
115     private static final String studFirstname = "Alain";
116     private static final String studName = "Tchana";
117
118     public static void main(String[] args) {
119         ResteasyClient client = new ResteasyClientBuilderImpl().build();
120         ResteasyWebTarget target = client.target(UriBuilder.fromPath(url));
121         ServiceInterface proxy = target.proxy(ServiceInterface.class);
122         Student student = proxy.getStudent(studFirstname, studName);
123         Record record = proxy.getRecord(student.getINE());
124         System.out.println("middleware grade of " + studFirstname + " "
125                             + studName + " = " + record.getMiddleware());
126     }
127 }
128
129 // Partie serveur
130
131 package n7.students_server;
132 import org.springframework.boot.builder.SpringApplicationBuilder;
133 import org.springframework.boot.web.servlet.support.SpringBootServletInitializer;
134
135 public class ServletInitializer extends SpringBootServletInitializer {
136     @Override
137     protected SpringApplicationBuilder configure(SpringApplicationBuilder application) {
138         return application.sources(StudentsServerApplication.class);
139     }
140 }
141
142 package n7.students_server;
143 import java.util.Hashtable;
144 import org.springframework.web.bind.annotation.GetMapping;
145 import org.springframework.web.bind.annotation.RequestParam;
146 import org.springframework.web.bind.annotation.RestController;
147
148 @RestController
149 public class StudentsController {
150     static Hashtable<String, Student> students = new Hashtable<String, Student>();
151     static Hashtable<String, Record> records = new Hashtable<String, Record>();
152
153     static void addStudent(Student s) {
154         students.put(s.getFirstname() + "-" + s.getLastname(), s);
155     }
156
157     static void addRecord(Record r) {
158         records.put(r.getINE(), r);
159     }
160
161     @GetMapping("/getstudent")

```

```

162     public Student getStudent(@RequestParam("firstname") String firstname,
163                               @RequestParam("lastname") String lastname) {
164         String key = firstname + "-" + lastname;
165         System.out.println("access student: " + key);
166         return students.get(key);
167     }
168
169     @GetMapping("/getrecord")
170     public Record getRecord(@RequestParam("ine") String ine) {
171         System.out.println("access record: " + ine);
172         return records.get(ine);
173     }
174
175     // Static initializer : bloc run une seule fois avant que le main soit appelé
176     static {
177         addStudent(new Student(...));
178         // Plusieurs addStudent ici avec des données fictives
179         addRecord(new Record(...));
180         // Plusieurs addRecord ici avec des données fictives
181     }
182 }
183
184 package n7.students_server;
185 import org.springframework.boot.SpringApplication;
186 import org.springframework.boot.autoconfigure.SpringBootApplication;
187
188 @SpringBootApplication
189 public class StudentsServerApplication {
190     public static void main(String[] args) {
191         SpringApplication.run(StudentsServerApplication.class, args);
192     }
193 }

```